

Literature Benchmark Brief: Jonkman_2009_NREL

Title: Definition of a 5-MW reference wind turbine for offshore system development

Authors: J. Jonkman, S. Butterfield, W. Musial, G. Scott (2009)

Venue: NREL Technical Report | **DOI:** 10.2172/947422

Cluster: Cluster C: Aerodynamics & Control Cliff

Core Theoretical Mechanism & Model Formulation

NREL 5-MW baseline specifications, rated aerodynamic torque, and blade-pitch curves

Empirical Scale & Experimental Dataset

Global gold-standard aeroelastic research turbine

Relevance to IEEE TSTE & Sustainable Energy Systems

Baseline aeroelastic parameters used across academic and industrial literature

Identified Gaps & Operational Vulnerabilities

Single-turbine aeroelastic model without wind plant spatial wake network

Role in Manuscript Defense & Reviewer #2 Proof

Benchmark reference for turbine aerodynamics and mechanical pitch rate limits